Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Diamond S. Ranch, Inc.

Physical address of dairy:

Number and Street	City	County	Zip Code
600 Roberts Ferry RD	Waterford	Stanislaus	95386

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1935

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X008-X012-X011-XXXX

B. OPERATORS

Sawyer, Chris

Operator name: Sawyer, Chris

Telephone no.: (209) 874-2312 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
600 Roberts Ferry RD	Waterford	CA	95386

C. OWNERS

Tom Sawyer, Family Trust

Legal owner name: Tom Sawyer, Family Trust

Telephone no.: (209) 874-2312 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
600 Roberts Ferry RD	Waterford	CA	95386

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	375	290	125	0
Number under roof	900	140	0	0	0	0
Maximum number	900	140	375	290	125	0
Average number	900	140	375	290	125	0
Avg live weight (lbs)	1,400	1,450	1,000	700		

Predominant milk cow breed: Holstein

Average milk production: 78 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 33,081.24 *tons per reporting period*

Total nitrogen from manure: 407,936.20 *lbs per reporting period*

After ammonia losses (30% loss applied): 285,555.34 *lbs per reporting period*

Total phosphorus from manure: 67,635.68 *lbs per reporting period*

Total potassium from manure: 177,088.70 *lbs per reporting period*

Total salt from manure: 455,958.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 24,384,000 *gallons*

Total nitrogen generated: 59,509.60 *lbs*

Total phosphorus generated: 6,354.76 *lbs*

Total potassium generated: 89,088.15 *lbs*

Total salt generated: 505,482.69 *lbs*

	24,384,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	24,384,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
IW Diamond S	Ground water
IW Pamona	Ground water
IW Robert	Ground water
River	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
05/19/2023	Solid commercial fertilizer *4-0-0*	25.80 *ton*	Dry-weight	0.1	4.000000	0.000000	0.000000	0.000000
06/08/2023	Solid commercial fertilizer *UN 32*	5.25 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	5,418.58	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	5,418.58	0.00	0.00	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Almond	100	91	1	manure	X008-X012-X011-XXXX
Bridge	6	6	2	manure	X008-X011-X013-XXXX
Bull Field	24	24	2	manure	X008-X011-X013-XXXX
Dryland 1	30	30	1	manure	X008-X012-X011-XXXX
Encino	36	36	2	process wastewater	X008-X012-X011-XXXX
Indian	13	13	2	manure	X008-X011-X013-XXXX
Machinery	47	47	2	process wastewater	X008-X012-X011-XXXX
No. 4	22	22	2	manure	X008-X012-X011-XXXX
Pamona	40	40	2	manure	X008-X012-X011-XXXX
Roberts Ferry	40	40	2	process wastewater	X008-X012-X011-XXXX
Upper Pamona	30	30	2	process wastewater	X008-X012-X011-XXXX
Vines	105	105	1	manure	X008-X012-X011-XXXX
Totals for areas that were used for application	493	484	21		
Totals for areas that were not used for application					
Land application area totals	493	484	21		

B. CROPS AND HARVESTS

Almond

Field name: Almond

02/02/2015: Almond Field Run

Crop: Almond Field Run Acres planted: 91 Plant date: 02/02/2015

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/17/2023	570.30 *ton*	As-is		8.7	13,800.00	2,380.00	37,000.00		44.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.50	279.50	47.45	302.25	0.00
Total actual harvest content	6.27	172.97	29.83	463.76	5,058.07

Bridge

Bridge

Field name: Bridge

10/19/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 6 Plant date: 10/19/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	102.00 *ton*	As-is		74.6	4,070.00	590.00	3,300.00		8.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.50	165.00	26.40	136.95	0.00
Total actual harvest content	17.00	138.38	20.06	112.20	751.33

05/19/2023: Corn, silage

Crop: Corn, silage Acres planted: 6 Plant date: 05/19/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/12/2023	134.20 *ton*	As-is		54.2	3,940.00	1,010.00	4,170.00		6.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	178.20	0.00
Total actual harvest content	22.37	176.25	45.18	186.54	1,270.25

Bull Field

Field name: Bull Field

10/21/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 24 Plant date: 10/21/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	396.00 *ton*	As-is		72.2	4,370.00	580.00	2,950.00		7.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.50	165.00	26.40	136.95	0.00
Total actual harvest content	16.50	144.21	19.14	97.35	651.35

Bull Field

05/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 24 Plant date: 05/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/12/2023	590.40 *ton*	As-is		52.6	3,740.00	1,320.00	3,760.00		4.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	178.20	0.00
Total actual harvest content	24.60	184.01	64.94	184.99	1,072.76

Dryland 1

Field name: Dryland 1

10/31/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 30 Plant date: 10/31/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	465.40 *ton*	As-is		63.6	3,840.00	880.00	3,960.00		6.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.50	165.00	26.40	136.95	0.00
Total actual harvest content	15.51	119.14	27.30	122.87	756.68

Encino

Field name: Encino

Encino

11/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 36 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	603.20 *ton*	As-is		78.6	3,660.00	530.00	2,040.00		6.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.50	165.00	26.40	136.95	0.00
Total actual harvest content	16.76	122.65	17.76	68.36	480.48

06/09/2023: Corn, silage

Crop: Corn, silage Acres planted: 36 Plant date: 06/09/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	946.80 *ton*	As-is		65.8	3,860.00	760.00	3,240.00		6.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	178.20	0.00
Total actual harvest content	26.30	203.04	39.98	170.42	1,097.34

Indian

Field name: Indian

10/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 13 Plant date: 10/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	214.50 *ton*	As-is		72.2	4,280.00	560.00	3,180.00		7.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.50	165.00	26.40	136.95	0.00
Total actual harvest content	16.50	141.24	18.48	104.94	724.75

Indian

05/23/2023: Corn, silage

Crop: Corn, silage Acres planted: 13 Plant date: 05/23/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/12/2023	331.50 *ton*	As-is		56.4	3,620.00	1,190.00	3,640.00		5.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	178.20	0.00
Total actual harvest content	25.50	184.62	60.69	185.64	1,156.27

Machinery

Field name: Machinery

10/24/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 47 Plant date: 10/24/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	754.50 *ton*	As-is		72.2	4,850.00	730.00	4,720.00		9.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.50	165.00	26.40	136.95	0.00
Total actual harvest content	16.05	155.72	23.44	151.54	883.63

06/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 47 Plant date: 06/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	1,224.00 *ton*	As-is		61.6	4,610.00	1,060.00	6,780.00		6.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	178.20	0.00
Total actual harvest content	26.04	240.11	55.21	353.14	1,340.05

No. 4

Field name: No. 4

10/25/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 22 Plant date: 10/25/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	342.60 *ton*	As-is		73.4	4,680.00	680.00	3,140.00		8.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.50	165.00	26.40	136.95	0.00
Total actual harvest content	15.57	145.76	21.18	97.80	671.06

06/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 22 Plant date: 06/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/05/2023	583.00 *ton*	As-is		64.6	3,680.00	720.00	2,970.00		4.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	178.20	0.00
Total actual harvest content	26.50	195.04	38.16	157.41	750.48

Pamona

Field name: Pamona

10/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 10/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	648.30 *ton*	As-is		71.6	3,920.00	660.00	2,870.00		7.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.50	165.00	26.40	136.95	0.00
Total actual harvest content	16.21	127.07	21.39	93.03	708.85

Pamona

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/05/2023	1,072.00 *ton*	As-is		67.6	3,300.00	560.00	3,220.00		5.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	178.20	0.00
Total actual harvest content	26.80	176.88	30.02	172.59	868.32

Roberts Ferry

Field name: Roberts Ferry

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	644.30 *ton*	As-is		69.4	3,390.00	620.00	3,500.00		7.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.50	165.00	26.40	136.95	0.00
Total actual harvest content	16.11	109.21	19.97	112.75	719.62

05/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 05/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	1,237.00 *ton*	As-is		68.4	3,670.00	570.00	4,040.00		5.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	178.20	0.00
Total actual harvest content	30.93	226.99	35.25	249.87	1,153.13

Upper Pamona

Field name: Upper Pamona

10/27/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 30 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	471.10 *ton*	As-is		55.6	4,460.00	890.00	4,540.00		6.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.50	165.00	26.40	136.95	0.00
Total actual harvest content	15.70	140.07	27.95	142.59	892.45

06/11/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 06/11/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	831.20 *ton*	As-is		71.4	3,320.00	620.00	2,490.00		5.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	178.20	0.00
Total actual harvest content	27.71	183.97	34.36	137.98	887.50

Vines

Field name: Vines

02/07/2014: Grape

Crop: Grape Acres planted: 105 Plant date: 02/07/2014

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/03/2023	1,271.00 *ton*	As-is		70.4	2,690.00	380.00	2,380.00		18.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	10.00	83.00	13.00	91.00	0.00
Total actual harvest content	12.10	65.12	9.20	57.62	1,332.88

NUTRIENT BUDGET

A. LAND APPLICATIONS

Almond - 02/02/2015: Almond Field Run

Field name: Almond

Crop: Almond Field Run

Plant date: 02/02/2015

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/28/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	58.16	12.97	22.74	2,315.03	220.00 *ton*
Application event totals		58.16	12.97	22.74	2,315.03	

03/03/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN17	Solid commercial fertilizer	51.00	0.00	0.00	0.00	
Application event totals		51.00	0.00	0.00	0.00	

04/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Diamond S	Ground water	0.00	0.00	0.00	111.48	9,209,200.00 *gal*
Application event totals		0.00	0.00	0.00	111.48	

04/15/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	48.00	0.00	0.00	0.00	
Application event totals		48.00	0.00	0.00	0.00	

04/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Diamond S	Ground water	0.00	0.00	0.00	111.48	9,209,200.00 *gal*
Application event totals		0.00	0.00	0.00	111.48	

Almond - 02/02/2015: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Diamond S	Ground water	0.00	0.00	0.00	111.37	9,200,100.00 *gal*
Application event totals		0.00	0.00	0.00	111.37	

05/13/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	48.00	0.00	0.00	0.00	
Application event totals		48.00	0.00	0.00	0.00	

05/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Diamond S	Ground water	0.00	0.00	0.00	111.37	9,200,100.00 *gal*
Application event totals		0.00	0.00	0.00	111.37	

06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Diamond S	Ground water	0.00	0.00	0.00	112.91	9,327,500.00 *gal*
Application event totals		0.00	0.00	0.00	112.91	

06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Diamond S	Ground water	0.00	0.00	0.00	112.91	9,327,500.00 *gal*
Application event totals		0.00	0.00	0.00	112.91	

07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Diamond S	Ground water	0.00	0.00	0.00	112.91	9,327,500.00 *gal*
Application event totals		0.00	0.00	0.00	112.91	

Almond - 02/02/2015: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Diamond S	Ground water	0.00	0.00	0.00	112.91	9,327,500.00 *gal*
Application event totals		0.00	0.00	0.00	112.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Diamond S	Ground water	0.00	0.00	0.00	112.91	9,327,500.00 *gal*
Application event totals		0.00	0.00	0.00	112.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Diamond S	Ground water	0.00	0.00	0.00	112.91	9,327,500.00 *gal*
Application event totals		0.00	0.00	0.00	112.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Diamond S	Ground water	0.00	0.00	0.00	112.91	9,327,500.00 *gal*
Application event totals		0.00	0.00	0.00	112.91	

Bridge - 10/19/2022: Oats, silage-soft dough

Field name: Bridge

Crop: Oats, silage-soft dough

Plant date: 10/19/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	591.45	787,500.00 *gal*
Application event totals		0.00	0.00	0.00	591.45	

Bridge - 10/19/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Settling Basin	Corral solids	184.39	46.42	88.56	5,495.42	50.00 *ton*
Application event totals		184.39	46.42	88.56	5,495.42	

Bridge - 05/19/2023: Corn, silage

Field name: Bridge

Crop: Corn, silage

Plant date: 05/19/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	65.42	735,000.00 *gal*
Application event totals		0.00	0.00	0.00	65.42	

04/30/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	191.00	40.50	99.80	4,800.00	60.00 *ton*
Application event totals		191.00	40.50	99.80	4,800.00	

05/19/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	60.75	682,500.00 *gal*
Application event totals		0.00	0.00	0.00	60.75	

Bridge - 05/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	58.42	656,250.00 *gal*
Application event totals		0.00	0.00	0.00	58.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	65.42	735,000.00 *gal*
Application event totals		0.00	0.00	0.00	65.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	60.75	682,500.00 *gal*
Application event totals		0.00	0.00	0.00	60.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	63.09	708,750.00 *gal*
Application event totals		0.00	0.00	0.00	63.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	60.75	682,500.00 *gal*
Application event totals		0.00	0.00	0.00	60.75	

Bridge - 05/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	58.42	656,250.00 *gal*
Application event totals		0.00	0.00	0.00	58.42	

Bull Field - 10/21/2022: Oats, silage-soft dough

Field name: Bull Field

Crop: Oats, silage-soft dough

Plant date: 10/21/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	571.74	3,045,000.00 *gal*
Application event totals		0.00	0.00	0.00	571.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Settling Basin	Corral solids	184.39	46.42	88.56	5,495.42	200.00 *ton*
Application event totals		184.39	46.42	88.56	5,495.42	

Bull Field - 05/20/2023: Corn, silage

Field name: Bull Field

Crop: Corn, silage

Plant date: 05/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Bull Field - 05/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	70.10	3,150,000.00 *gal*
Application event totals		0.00	0.00	0.00	70.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	198.96	42.19	103.96	5,000.00	250.00 *ton*
Application event totals		198.96	42.19	103.96	5,000.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	68.93	3,097,500.00 *gal*
Application event totals		0.00	0.00	0.00	68.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	67.76	3,045,000.00 *gal*
Application event totals		0.00	0.00	0.00	67.76	

Bull Field - 05/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	65.42	2,940,000.00 *gal*
Application event totals		0.00	0.00	0.00	65.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	66.59	2,992,500.00 *gal*
Application event totals		0.00	0.00	0.00	66.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	63.09	2,835,000.00 *gal*
Application event totals		0.00	0.00	0.00	63.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	65.42	2,940,000.00 *gal*
Application event totals		0.00	0.00	0.00	65.42	

Dryland 1 - 10/31/2022: Oats, silage-soft dough

Field name: Dryland 1

Crop: Oats, silage-soft dough

Plant date: 10/31/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	154.89	38.99	74.39	4,616.15	210.00 *ton*
Application event totals		154.89	38.99	74.39	4,616.15	

Encino - 11/04/2022: Oats, silage-soft dough

Field name: Encino

Crop: Oats, silage-soft dough

Plant date: 11/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/14/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	72.03	3.82	172.42	1,330.75	1,248,000.00 *gal*
River	Ground water	0.00	0.00	0.00	351.49	2,808,000.00 *gal*
Application event totals		72.03	3.82	172.42	1,682.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	92.63	13.16	149.38	851.19	1,200,000.00 *gal*
Application event totals		92.63	13.16	149.38	851.19	

Encino - 06/09/2023: Corn, silage

Field name: Encino

Crop: Corn, silage

Plant date: 06/09/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	179.58	10.71	226.98	1,235.06	1,440,000.00 *gal*
River	Ground water	0.00	0.00	0.00	48.07	3,240,000.00 *gal*
Application event totals		179.58	10.71	226.98	1,283.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Encino - 06/09/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	67.29	4,536,000.00 *gal*
Application event totals		0.00	0.00	0.00	67.29	

07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	65.69	4,428,000.00 *gal*
Application event totals		0.00	0.00	0.00	65.69	

07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	83.90	9.55	100.03	438.84	1,392,000.00 *gal*
River	Ground water	0.00	0.00	0.00	46.46	3,132,000.00 *gal*
Application event totals		83.90	9.55	100.03	485.30	

08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	65.69	4,428,000.00 *gal*
Application event totals		0.00	0.00	0.00	65.69	

08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	64.09	4,320,000.00 *gal*
Application event totals		0.00	0.00	0.00	64.09	

09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	59.28	3,996,000.00 *gal*
Application event totals		0.00	0.00	0.00	59.28	

Indian - 10/20/2022: Oats, silage-soft dough

Field name: Indian

Crop: Oats, silage-soft dough

Plant date: 10/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	582.35	1,680,000.00 *gal*
Application event totals		0.00	0.00	0.00	582.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	170.21	42.85	81.75	5,072.69	100.00 *ton*
Application event totals		170.21	42.85	81.75	5,072.69	

Indian - 05/23/2023: Corn, silage

Field name: Indian

Crop: Corn, silage

Plant date: 05/23/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	191.00	40.50	99.80	4,800.00	130.00 *ton*
Application event totals		191.00	40.50	99.80	4,800.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	73.33	1,785,000.00 *gal*
Application event totals		0.00	0.00	0.00	73.33	

Indian - 05/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	69.02	1,680,000.00 *gal*
Application event totals		0.00	0.00	0.00	69.02	

06/15/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	69.02	1,680,000.00 *gal*
Application event totals		0.00	0.00	0.00	69.02	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	69.02	1,680,000.00 *gal*
Application event totals		0.00	0.00	0.00	69.02	

07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	69.02	1,680,000.00 *gal*
Application event totals		0.00	0.00	0.00	69.02	

Indian - 05/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	66.86	1,627,500.00 *gal*
Application event totals		0.00	0.00	0.00	66.86	

08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	64.71	1,575,000.00 *gal*
Application event totals		0.00	0.00	0.00	64.71	

Machinery - 10/24/2022: Oats, silage-soft dough

Field name: Machinery

Crop: Oats, silage-soft dough Plant date: 10/24/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	80.64	4.27	193.02	1,489.74	1,824,000.00 *gal*
IW Pamona	Ground water	10.75	0.00	0.00	148.77	3,420,000.00 *gal*
Application event totals		91.39	4.27	193.02	1,638.51	

02/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	119.20	16.93	192.22	1,095.32	2,016,000.00 *gal*
Application event totals		119.20	16.93	192.22	1,095.32	

Machinery - 06/08/2023: Corn, silage

Field name: Machinery

Crop: Corn, silage Plant date: 06/08/2023

Machinery - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Pamona	Ground water	14.54	0.00	0.00	738.27	6,300,000.00 *gal*
Application event totals		14.54	0.00	0.00	738.27	

06/08/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Pamona	Ground water	13.92	0.00	0.00	706.63	6,030,000.00 *gal*
Application event totals		13.92	0.00	0.00	706.63	

07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Pamona	Ground water	9.56	0.00	0.00	485.15	4,140,000.00 *gal*
Application event totals		9.56	0.00	0.00	485.15	

07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	101.93	11.60	121.53	533.17	2,208,000.00 *gal*
Application event totals		101.93	11.60	121.53	533.17	

07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Pamona	Ground water	13.92	0.00	0.00	706.63	6,030,000.00 *gal*
Application event totals		13.92	0.00	0.00	706.63	

Machinery - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	99.71	11.35	118.89	521.58	2,160,000.00 *gal*
IW Pamona	Ground water	9.35	0.00	0.00	474.60	4,050,000.00 *gal*
Application event totals		109.06	11.35	118.89	996.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Pamona	Ground water	13.50	0.00	0.00	685.53	5,850,000.00 *gal*
Application event totals		13.50	0.00	0.00	685.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Pamona	Ground water	13.09	0.00	0.00	664.44	5,670,000.00 *gal*
Application event totals		13.09	0.00	0.00	664.44	

No. 4 - 10/25/2022: Oats, silage-soft dough

Field name: No. 4

Crop: Oats, silage-soft dough

Plant date: 10/25/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	191.10	48.10	91.78	5,695.25	190.00 *ton*
Application event totals		191.10	48.10	91.78	5,695.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/04/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	553.05	2,700,000.00 *gal*
Application event totals		0.00	0.00	0.00	553.05	

No. 4 - 06/14/2023: Corn, silage

Field name: No. 4

Crop: Corn, silage

Plant date: 06/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	208.36	44.18	108.87	5,236.36	240.00 *ton*
Application event totals		208.36	44.18	108.87	5,236.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	73.41	3,024,000.00 *gal*
Application event totals		0.00	0.00	0.00	73.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	70.79	2,916,000.00 *gal*
Application event totals		0.00	0.00	0.00	70.79	

No. 4 - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	68.17	2,808,000.00 *gal*
Application event totals		0.00	0.00	0.00	68.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	68.17	2,808,000.00 *gal*
Application event totals		0.00	0.00	0.00	68.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	68.17	2,808,000.00 *gal*
Application event totals		0.00	0.00	0.00	68.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	68.17	2,808,000.00 *gal*
Application event totals		0.00	0.00	0.00	68.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	65.55	2,700,000.00 *gal*
Application event totals		0.00	0.00	0.00	65.55	

Pamona - 10/26/2022: Oats, silage-soft dough

Field name: Pamona

Crop: Oats, silage-soft dough

Plant date: 10/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Pamona - 10/26/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Settling Basin	Corral solids	177.02	44.56	85.02	5,275.60	320.00 *ton*
Application event totals		177.02	44.56	85.02	5,275.60	

10/05/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	547.52	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	547.52	

Pamona - 06/15/2023: Corn, silage

Field name: Pamona

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	200.55	42.53	104.79	5,040.00	420.00 *ton*
Application event totals		200.55	42.53	104.79	5,040.00	

05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	72.10	5,400,000.00 *gal*
Application event totals		0.00	0.00	0.00	72.10	

06/15/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Pamona - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	69.22	5,184,000.00 *gal*
Application event totals		0.00	0.00	0.00	69.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	69.22	5,184,000.00 *gal*
Application event totals		0.00	0.00	0.00	69.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	69.22	5,184,000.00 *gal*
Application event totals		0.00	0.00	0.00	69.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	67.77	5,076,000.00 *gal*
Application event totals		0.00	0.00	0.00	67.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	67.77	5,076,000.00 *gal*
Application event totals		0.00	0.00	0.00	67.77	

Pamona - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Ground water	0.00	0.00	0.00	66.33	4,968,000.00 *gal*
Application event totals		0.00	0.00	0.00	66.33	

Roberts Ferry - 11/03/2022: Oats, silage-soft dough

Field name: Roberts Ferry

Crop: Oats, silage-soft dough

Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/13/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Robert	Ground water	3.94	0.00	0.00	377.87	4,830,000.00 *gal*
Application event totals		3.94	0.00	0.00	377.87	

02/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	120.05	17.05	193.59	1,103.14	1,728,000.00 *gal*
Application event totals		120.05	17.05	193.59	1,103.14	

Roberts Ferry - 05/20/2023: Corn, silage

Field name: Roberts Ferry

Crop: Corn, silage

Plant date: 05/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Roberts Ferry - 05/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Robert	Ground water	4.52	0.00	0.00	523.23	5,382,000.00 *gal*
Application event totals		4.52	0.00	0.00	523.23	

05/20/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Robert	Ground water	4.47	0.00	0.00	516.53	5,313,000.00 *gal*
Application event totals		4.47	0.00	0.00	516.53	

06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Robert	Ground water	4.41	0.00	0.00	509.82	5,244,000.00 *gal*
Application event totals		4.41	0.00	0.00	509.82	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	119.77	13.64	142.80	626.48	2,208,000.00 *gal*
IW Robert	Ground water	2.67	0.00	0.00	308.57	3,174,000.00 *gal*
Application event totals		122.44	13.64	142.80	935.05	

07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	117.16	13.34	139.70	612.86	2,160,000.00 *gal*
IW Robert	Ground water	2.61	0.00	0.00	301.87	3,105,000.00 *gal*
Application event totals		119.77	13.34	139.70	914.72	

Roberts Ferry - 05/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Robert	Ground water	4.35	0.00	0.00	503.11	5,175,000.00 *gal*
Application event totals		4.35	0.00	0.00	503.11	

08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Robert	Ground water	4.29	0.00	0.00	496.40	5,106,000.00 *gal*
Application event totals		4.29	0.00	0.00	496.40	

Upper Pamona - 10/27/2022: Oats, silage-soft dough

Field name: Upper Pamona

Crop: Oats, silage-soft dough Plant date: 10/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	83.12	4.41	198.94	1,535.48	1,200,000.00 *gal*
IW Pamona	Ground water	11.08	0.00	0.00	153.34	2,250,000.00 *gal*
Application event totals		94.19	4.41	198.94	1,688.82	

02/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	97.82	13.89	157.74	898.86	1,056,000.00 *gal*
Application event totals		97.82	13.89	157.74	898.86	

Upper Pamona - 06/11/2023: Corn, silage

Field name: Upper Pamona

Crop: Corn, silage Plant date: 06/11/2023

Upper Pamona - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Pamona	Ground water	14.65	0.00	0.00	743.54	4,050,000.00 *gal*
Application event totals		14.65	0.00	0.00	743.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Pamona	Ground water	14.32	0.00	0.00	727.02	3,960,000.00 *gal*
Application event totals		14.32	0.00	0.00	727.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Pamona	Ground water	8.79	0.00	0.00	446.12	2,430,000.00 *gal*
Application event totals		8.79	0.00	0.00	446.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	93.73	10.67	111.76	490.29	1,296,000.00 *gal*
Application event totals		93.73	10.67	111.76	490.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Pamona	Ground water	14.32	0.00	0.00	727.02	3,960,000.00 *gal*
Application event totals		14.32	0.00	0.00	727.02	

Upper Pamona - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	90.26	10.28	107.62	472.13	1,248,000.00 *gal*
IW Pamona	Ground water	8.46	0.00	0.00	429.60	2,340,000.00 *gal*
Application event totals		98.72	10.28	107.62	901.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Pamona	Ground water	13.67	0.00	0.00	693.97	3,780,000.00 *gal*
Application event totals		13.67	0.00	0.00	693.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Pamona	Ground water	13.02	0.00	0.00	660.92	3,600,000.00 *gal*
Application event totals		13.02	0.00	0.00	660.92	

Vines - 02/07/2014: Grape

Field name: Vines

Crop: Grape

Plant date: 02/07/2014

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/30/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	45.82	10.22	17.92	1,823.97	200.00 *ton*
Application event totals		45.82	10.22	17.92	1,823.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Diamond S	Ground water	0.00	0.00	0.00	88.29	8,415,750.00 *gal*
Application event totals		0.00	0.00	0.00	88.29	

Vines - 02/07/2014: Grape

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/18/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	9.60	0.00	0.00	0.00	
Application event totals		9.60	0.00	0.00	0.00	

05/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Diamond S	Ground water	0.00	0.00	0.00	88.29	8,415,750.00 *gal*
Application event totals		0.00	0.00	0.00	88.29	

05/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Diamond S	Ground water	0.00	0.00	0.00	88.29	8,415,750.00 *gal*
Application event totals		0.00	0.00	0.00	88.29	

05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Diamond S	Ground water	0.00	0.00	0.00	88.29	8,415,750.00 *gal*
Application event totals		0.00	0.00	0.00	88.29	

05/30/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	9.60	0.00	0.00	0.00	
Application event totals		9.60	0.00	0.00	0.00	

06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Diamond S	Ground water	0.00	0.00	0.00	88.29	8,415,750.00 *gal*
Application event totals		0.00	0.00	0.00	88.29	

Vines - 02/07/2014: Grape

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Diamond S	Ground water	0.00	0.00	0.00	90.88	8,662,500.00 *gal*
Application event totals		0.00	0.00	0.00	90.88	

07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Diamond S	Ground water	0.00	0.00	0.00	90.88	8,662,500.00 *gal*
Application event totals		0.00	0.00	0.00	90.88	

07/11/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	9.60	0.00	0.00	0.00	
Application event totals		9.60	0.00	0.00	0.00	

07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Diamond S	Ground water	0.00	0.00	0.00	90.88	8,662,500.00 *gal*
Application event totals		0.00	0.00	0.00	90.88	

08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Diamond S	Ground water	0.00	0.00	0.00	90.88	8,662,500.00 *gal*
Application event totals		0.00	0.00	0.00	90.88	

08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Diamond S	Ground water	0.00	0.00	0.00	90.88	8,662,500.00 *gal*
Application event totals		0.00	0.00	0.00	90.88	

Vines - 02/07/2014: Grape

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Diamond S	Ground water	0.00	0.00	0.00	90.88	8,662,500.00 *gal*
Application event totals		0.00	0.00	0.00	90.88	

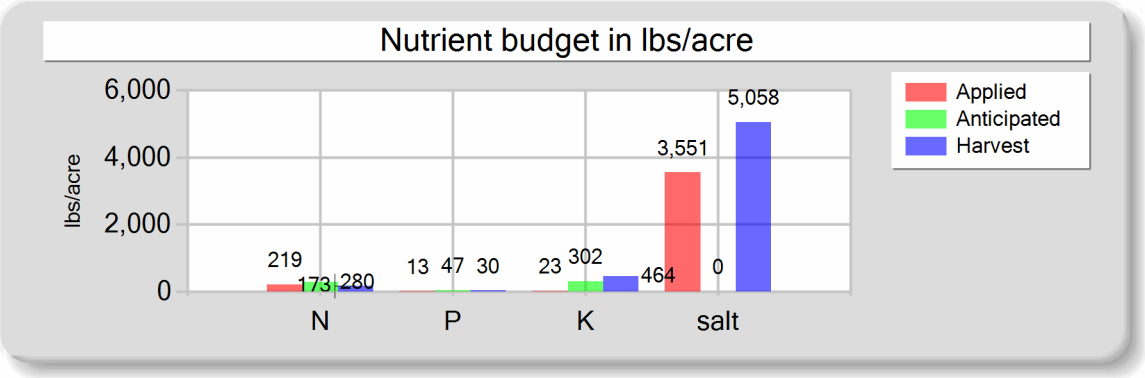
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Diamond S	Ground water	0.00	0.00	0.00	90.88	8,662,500.00 *gal*
Application event totals		0.00	0.00	0.00	90.88	

B. NUTRIENT BUDGET

Almond - 02/02/2015: Almond Field Run

Field name: Almond | Crop: Almond Field Run | Plant date: 02/02/2015



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	147.00	0.00	0.00	0.00
Dry manure	58.16	12.97	22.74	2,315.03
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,236.04
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	219.16	12.97	22.74	3,551.07
Anticipated crop nutrient removal	279.50	47.45	302.25	0.00
Actual crop nutrient removal	172.97	29.83	463.76	5,058.07
Nutrient balance	46.19	-16.86	-441.02	-1,507.00
Applied to removed ratio	1.27	0.43	0.05	0.70

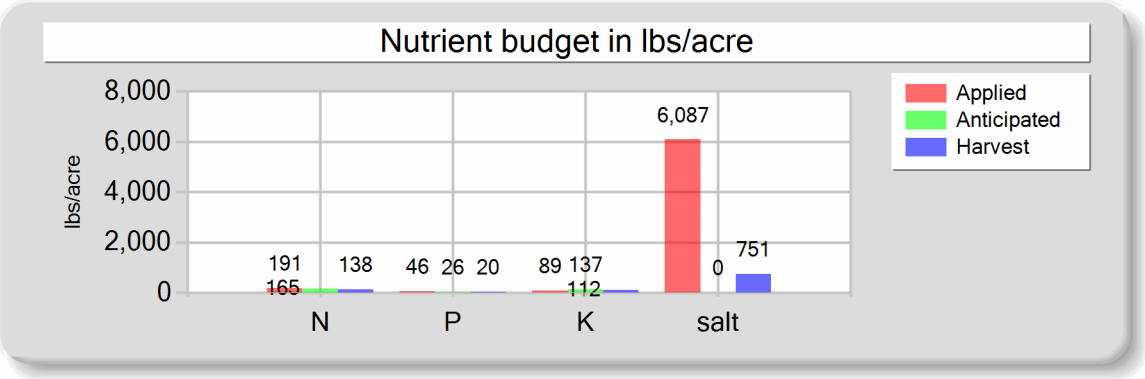
Fresh water applied	
102,111,100.00	*gallons*
3,760.40	*acre-inches*
41.32	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Bridge - 10/19/2022: Oats, silage-soft dough

Field name: Bridge | Crop: Oats, silage-soft dough | Plant date: 10/19/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	184.39	46.42	88.56	5,495.42
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	591.45
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	191.39	46.42	88.56	6,086.87
Anticipated crop nutrient removal	165.00	26.40	136.95	0.00
Actual crop nutrient removal	138.38	20.06	112.20	751.33
Nutrient balance	53.01	26.36	-23.64	5,335.54
Applied to removed ratio	1.38	2.31	0.79	8.10

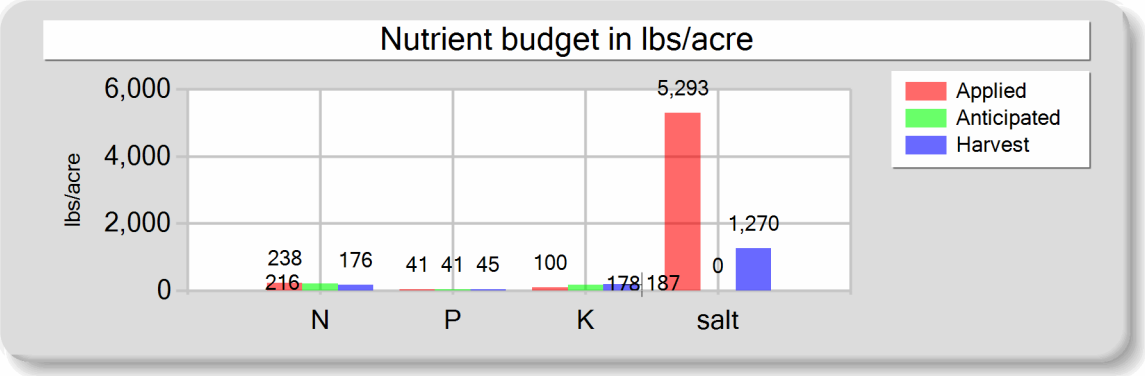
Fresh water applied	
787,500.00	*gallons*
29.00	*acre-inches*
4.83	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Bridge - 05/19/2023: Corn, silage

Field name: Bridge | Crop: Corn, silage | Plant date: 05/19/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	40.00	0.00	0.00	0.00
Dry manure	191.00	40.50	99.80	4,800.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	493.02
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	238.00	40.50	99.80	5,293.02
Anticipated crop nutrient removal	216.00	40.50	178.20	0.00
Actual crop nutrient removal	176.25	45.18	186.54	1,270.25
Nutrient balance	61.75	-4.68	-86.74	4,022.77
Applied to removed ratio	1.35	0.90	0.54	4.17

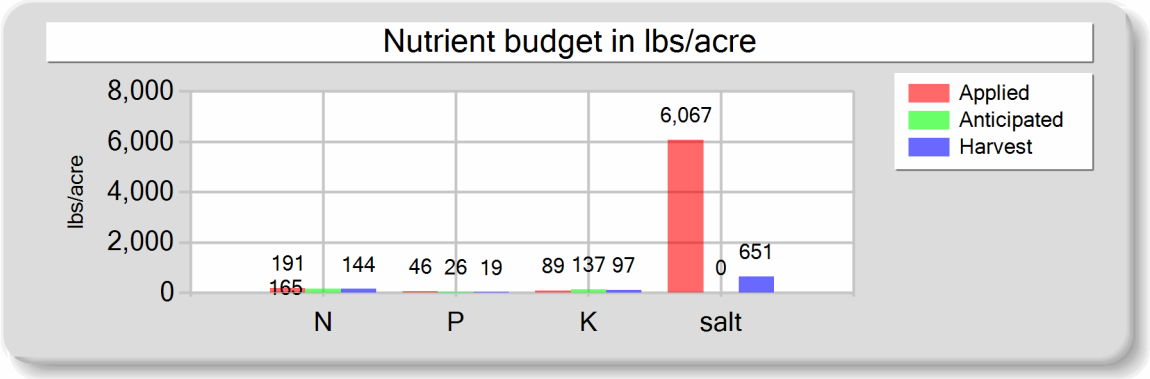
Fresh water applied	
5,538,750.00	*gallons*
203.97	*acre-inches*
34.00	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Bull Field - 10/21/2022: Oats, silage-soft dough

Field name: Bull Field | Crop: Oats, silage-soft dough | Plant date: 10/21/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	184.39	46.42	88.56	5,495.42
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	571.74
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	191.39	46.42	88.56	6,067.15
Anticipated crop nutrient removal	165.00	26.40	136.95	0.00
Actual crop nutrient removal	144.21	19.14	97.35	651.35
Nutrient balance	47.18	27.28	-8.79	5,415.80
Applied to removed ratio	1.33	2.43	0.91	9.31

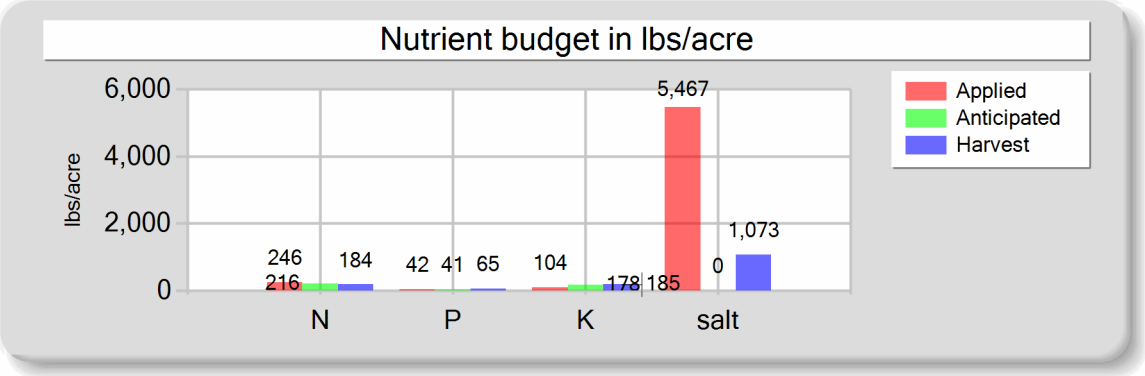
Fresh water applied	
3,045,000.00	*gallons*
112.14	*acre-inches*
4.67	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Bull Field - 05/20/2023: Corn, silage

Field name: Bull Field Crop: Corn, silage Plant date: 05/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	40.00	0.00	0.00	0.00
Dry manure	198.96	42.19	103.96	5,000.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	467.32
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	245.96	42.19	103.96	5,467.32
Anticipated crop nutrient removal	216.00	40.50	178.20	0.00
Actual crop nutrient removal	184.01	64.94	184.99	1,072.76
Nutrient balance	61.95	-22.76	-81.03	4,394.56
Applied to removed ratio	1.34	0.65	0.56	5.10

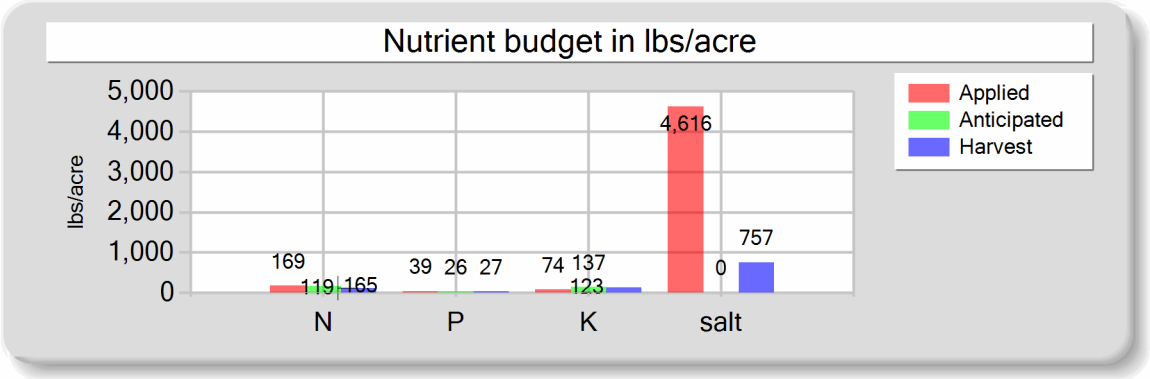
Fresh water applied	
21,000,000.00	*gallons*
773.36	*acre-inches*
32.22	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dryland 1 - 10/31/2022: Oats, silage-soft dough

Field name: Dryland 1 Crop: Oats, silage-soft dough Plant date: 10/31/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	154.89	38.99	74.39	4,616.15
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	168.89	38.99	74.39	4,616.15
Anticipated crop nutrient removal	165.00	26.40	136.95	0.00
Actual crop nutrient removal	119.14	27.30	122.87	756.68
Nutrient balance	49.75	11.69	-48.47	3,859.47
Applied to removed ratio	1.42	1.43	0.61	6.10

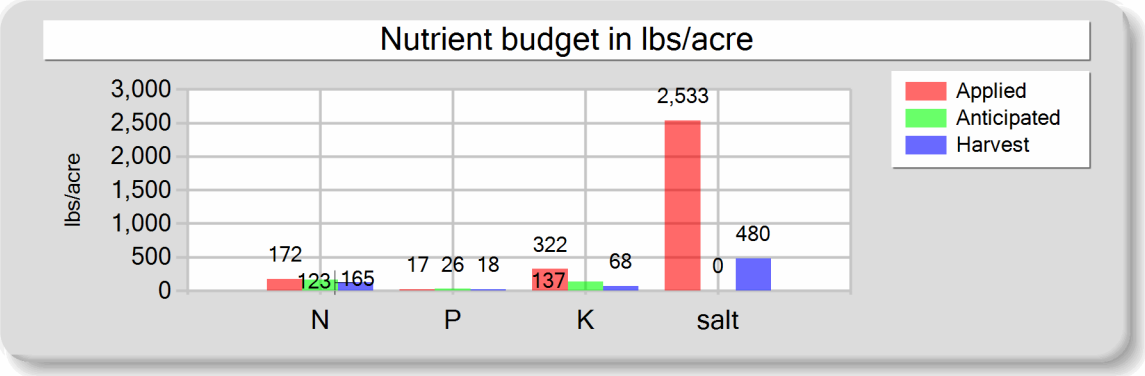
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Encino - 11/04/2022: Oats, silage-soft dough

Field name: Encino | Crop: Oats, silage-soft dough | Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	164.66	16.98	321.79	2,181.94
Fresh water	0.00	0.00	0.00	351.49
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	171.66	16.98	321.79	2,533.43
Anticipated crop nutrient removal	165.00	26.40	136.95	0.00
Actual crop nutrient removal	122.65	17.76	68.36	480.48
Nutrient balance	49.01	-0.78	253.43	2,052.95
Applied to removed ratio	1.40	0.96	4.71	5.27

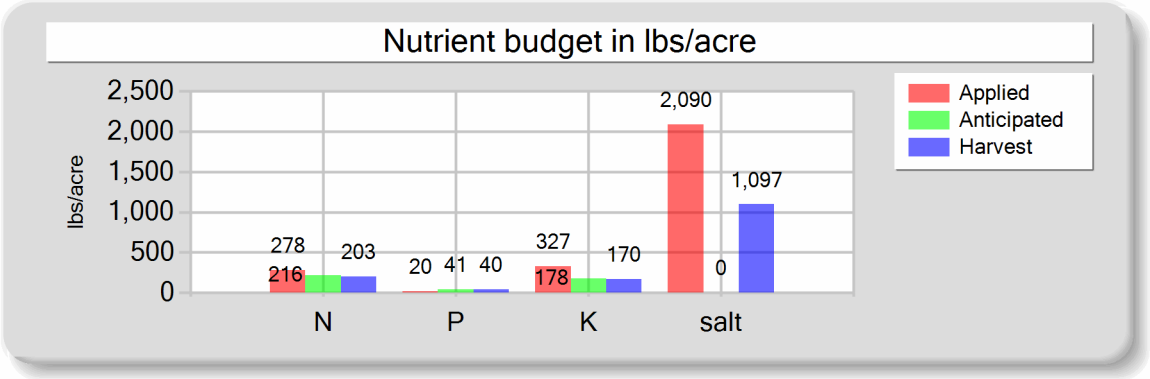
Fresh water applied	
2,808,000.00	*gallons*
103.41	*acre-inches*
2.87	*inches/acre*

Process wastewater applied	
2,448,000.00	*gallons*
90.15	*acre-inches*
2.50	*inches/acre*

Total harvests for the crop	
1	*harvests*

Encino - 06/09/2023: Corn, silage

Field name: Encino | Crop: Corn, silage | Plant date: 06/09/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	263.48	20.27	327.01	1,673.90
Fresh water	0.00	0.00	0.00	416.58
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	278.48	20.27	327.01	2,090.48
Anticipated crop nutrient removal	216.00	40.50	178.20	0.00
Actual crop nutrient removal	203.04	39.98	170.42	1,097.34
Nutrient balance	75.44	-19.71	156.59	993.14
Applied to removed ratio	1.37	0.51	1.92	1.91

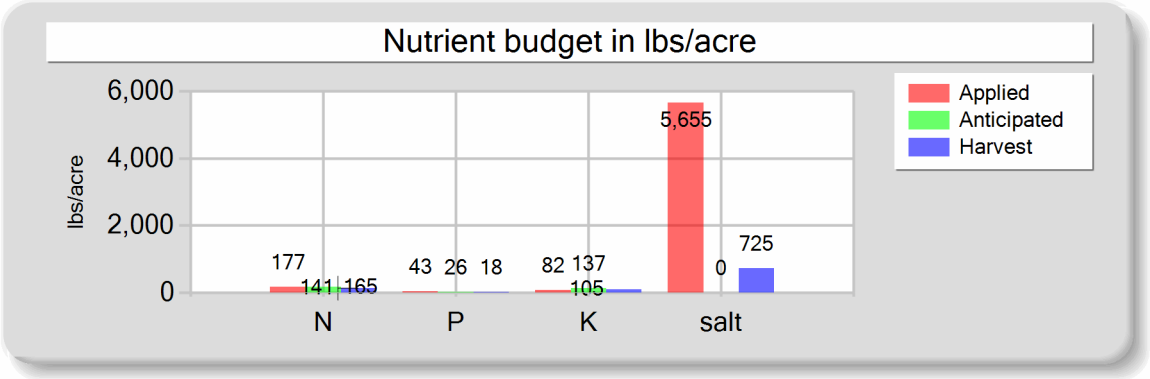
Fresh water applied	
28,080,000.00	*gallons*
1,034.09	*acre-inches*
28.72	*inches/acre*

Process wastewater applied	
2,832,000.00	*gallons*
104.29	*acre-inches*
2.90	*inches/acre*

Total harvests for the crop	
1	*harvests*

Indian - 10/20/2022: Oats, silage-soft dough

Field name: Indian | Crop: Oats, silage-soft dough | Plant date: 10/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	170.21	42.85	81.75	5,072.69
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	582.35
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	177.21	42.85	81.75	5,655.04
Anticipated crop nutrient removal	165.00	26.40	136.95	0.00
Actual crop nutrient removal	141.24	18.48	104.94	724.75
Nutrient balance	35.97	24.37	-23.19	4,930.30
Applied to removed ratio	1.25	2.32	0.78	7.80

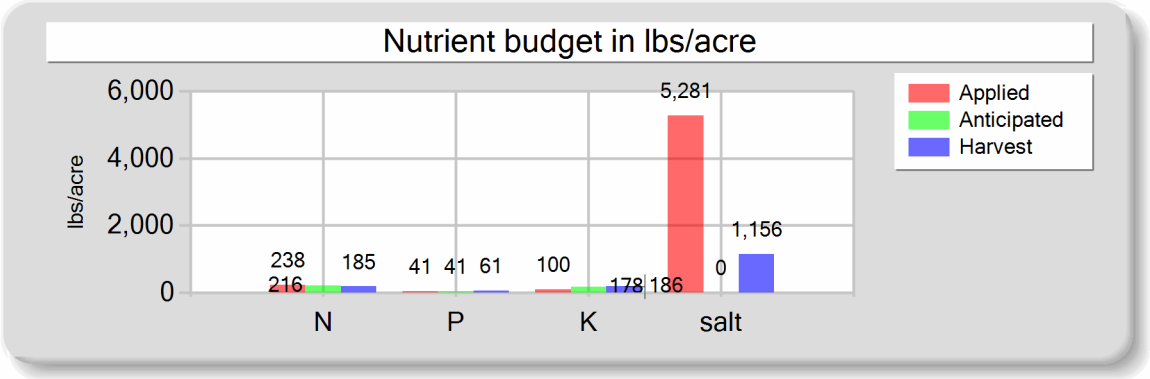
Fresh water applied	
1,680,000.00	*gallons*
61.87	*acre-inches*
4.76	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Indian - 05/23/2023: Corn, silage

Field name: Indian | Crop: Corn, silage | Plant date: 05/23/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	40.00	0.00	0.00	0.00
Dry manure	191.00	40.50	99.80	4,800.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	480.98
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	238.00	40.50	99.80	5,280.98
Anticipated crop nutrient removal	216.00	40.50	178.20	0.00
Actual crop nutrient removal	184.62	60.69	185.64	1,156.27
Nutrient balance	53.38	-20.19	-85.84	4,124.71
Applied to removed ratio	1.29	0.67	0.54	4.57

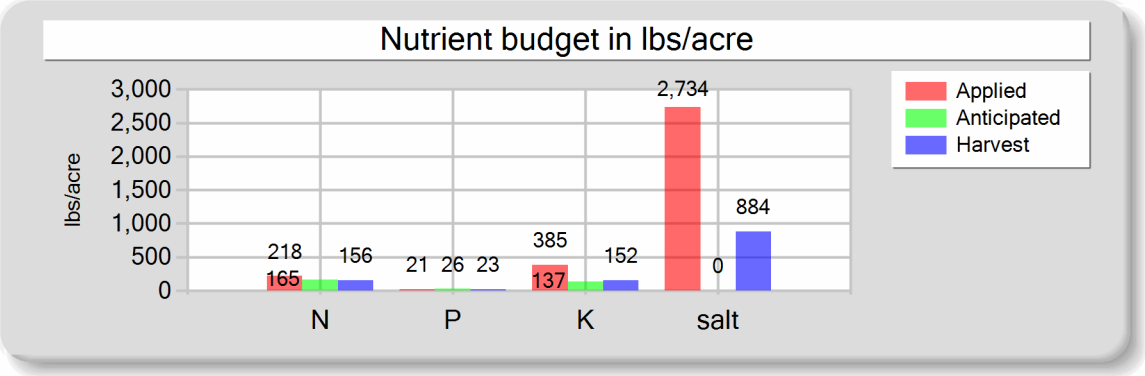
Fresh water applied	
11,707,500.00	*gallons*
431.15	*acre-inches*
33.17	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Machinery - 10/24/2022: Oats, silage-soft dough

Field name: Machinery | Crop: Oats, silage-soft dough | Plant date: 10/24/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	199.84	21.21	385.24	2,585.06
Fresh water	10.75	0.00	0.00	148.77
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	217.58	21.21	385.24	2,733.83
Anticipated crop nutrient removal	165.00	26.40	136.95	0.00
Actual crop nutrient removal	155.72	23.44	151.54	883.63
Nutrient balance	61.87	-2.23	233.69	1,850.20
Applied to removed ratio	1.40	0.90	2.54	3.09

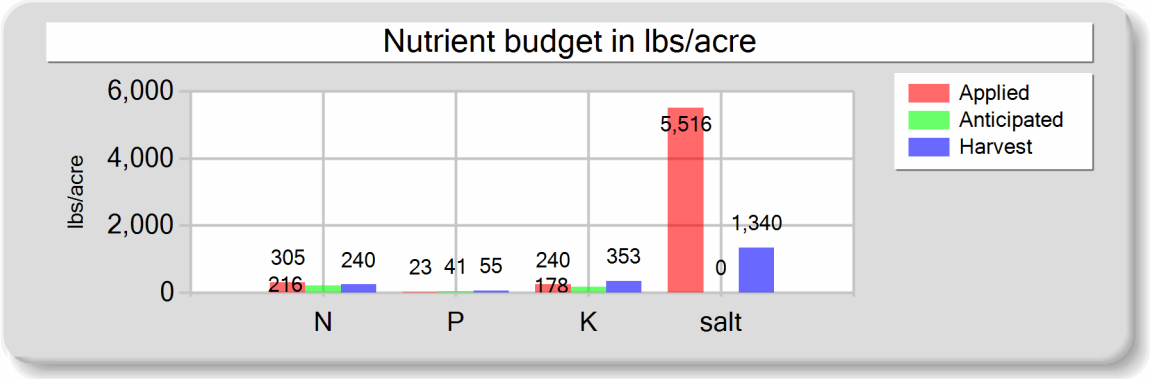
Fresh water applied	
3,420,000.00	*gallons*
125.95	*acre-inches*
2.68	*inches/acre*

Process wastewater applied	
3,840,000.00	*gallons*
141.41	*acre-inches*
3.01	*inches/acre*

Total harvests for the crop	
1	*harvests*

Machinery - 06/08/2023: Corn, silage

Field name: Machinery | Crop: Corn, silage | Plant date: 06/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	201.64	22.96	240.42	1,054.75
Fresh water	87.87	0.00	0.00	4,461.24
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	304.52	22.96	240.42	5,515.99
Anticipated crop nutrient removal	216.00	40.50	178.20	0.00
Actual crop nutrient removal	240.11	55.21	353.14	1,340.05
Nutrient balance	64.40	-32.25	-112.72	4,175.94
Applied to removed ratio	1.27	0.42	0.68	4.12

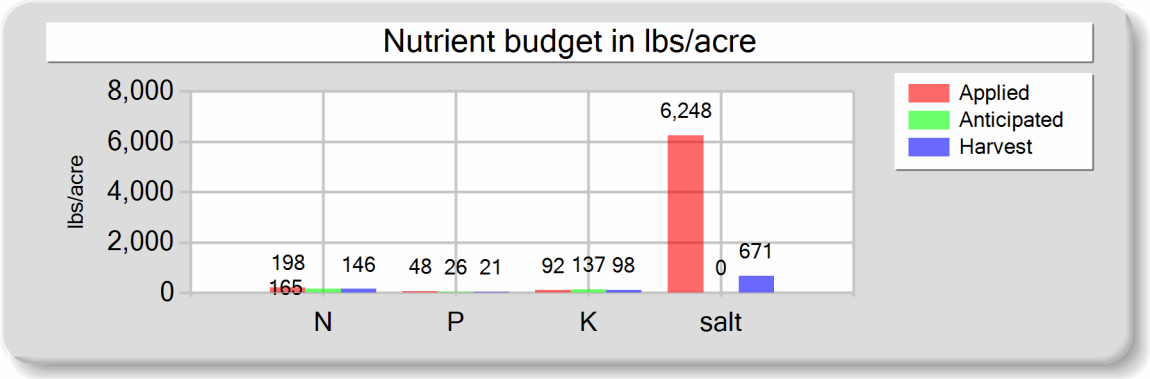
Fresh water applied	
38,070,000.00	*gallons*
1,401.99	*acre-inches*
29.83	*inches/acre*

Process wastewater applied	
4,368,000.00	*gallons*
160.86	*acre-inches*
3.42	*inches/acre*

Total harvests for the crop	
1	*harvests*

No. 4 - 10/25/2022: Oats, silage-soft dough

Field name: No. 4 Crop: Oats, silage-soft dough Plant date: 10/25/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	191.10	48.10	91.78	5,695.25
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	553.05
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	198.10	48.10	91.78	6,248.30
Anticipated crop nutrient removal	165.00	26.40	136.95	0.00
Actual crop nutrient removal	145.76	21.18	97.80	671.06
Nutrient balance	52.34	26.92	-6.01	5,577.24
Applied to removed ratio	1.36	2.27	0.94	9.31

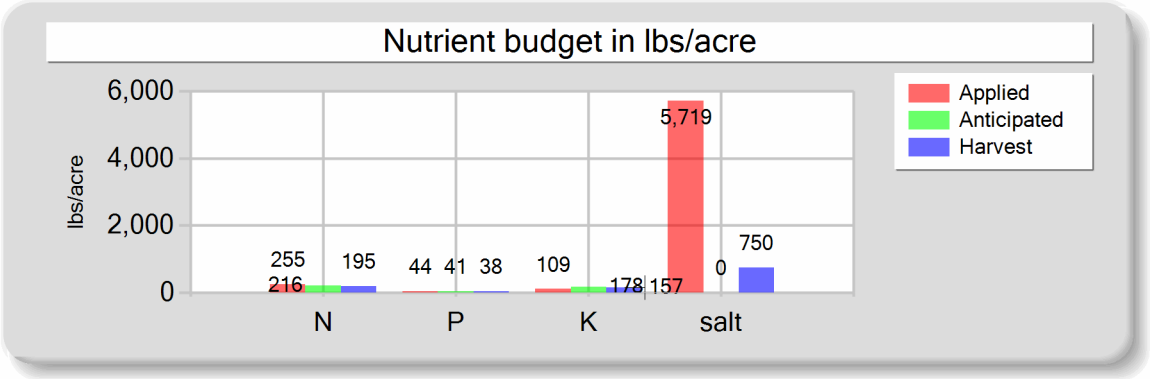
Fresh water applied
2,700,000.00 *gallons*
99.43 *acre-inches*
4.52 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

No. 4 - 06/14/2023: Corn, silage

Field name: No. 4 Crop: Corn, silage Plant date: 06/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	40.00	0.00	0.00	0.00
Dry manure	208.36	44.18	108.87	5,236.36
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	482.42
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	255.36	44.18	108.87	5,718.78
Anticipated crop nutrient removal	216.00	40.50	178.20	0.00
Actual crop nutrient removal	195.04	38.16	157.41	750.48
Nutrient balance	60.32	6.02	-48.54	4,968.30
Applied to removed ratio	1.31	1.16	0.69	7.62

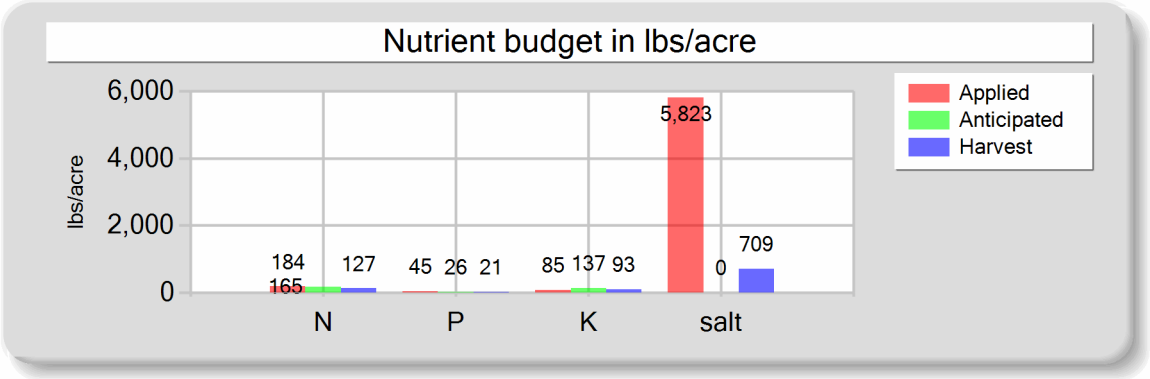
Fresh water applied	
19,872,000.00	*gallons*
731.82	*acre-inches*
33.26	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Pamona - 10/26/2022: Oats, silage-soft dough

Field name: Pamona | Crop: Oats, silage-soft dough | Plant date: 10/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	177.02	44.56	85.02	5,275.60
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	547.52
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	184.02	44.56	85.02	5,823.12
Anticipated crop nutrient removal	165.00	26.40	136.95	0.00
Actual crop nutrient removal	127.07	21.39	93.03	708.85
Nutrient balance	56.95	23.17	-8.01	5,114.26
Applied to removed ratio	1.45	2.08	0.91	8.21

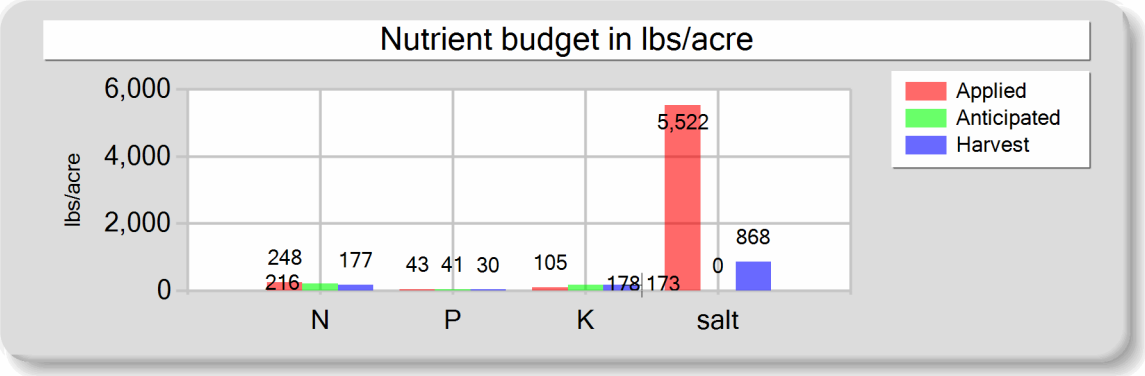
Fresh water applied	
4,860,000.00	*gallons*
178.98	*acre-inches*
4.47	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Pamona - 06/15/2023: Corn, silage

Field name: Pamona | Crop: Corn, silage | Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	40.00	0.00	0.00	0.00
Dry manure	200.55	42.53	104.79	5,040.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	481.63
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	247.55	42.53	104.79	5,521.63
Anticipated crop nutrient removal	216.00	40.50	178.20	0.00
Actual crop nutrient removal	176.88	30.02	172.59	868.32
Nutrient balance	70.67	12.51	-67.80	4,653.31
Applied to removed ratio	1.40	1.42	0.61	6.36

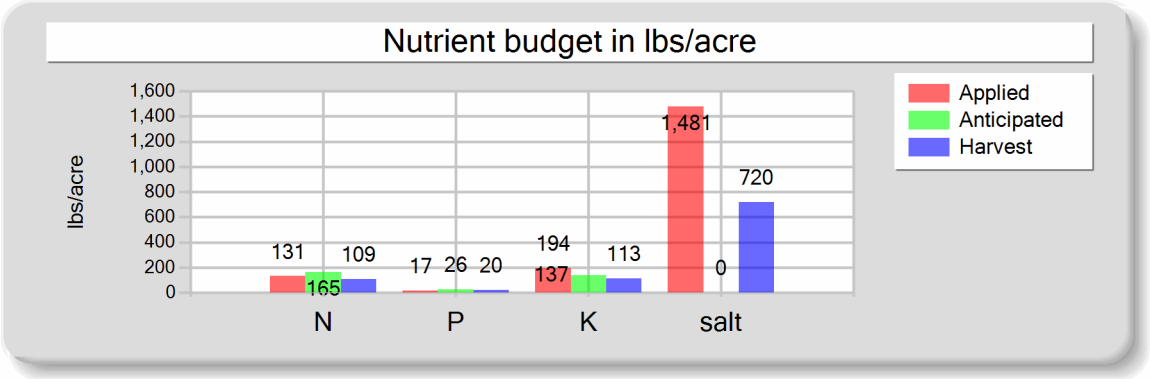
Fresh water applied	
36,072,000.00	*gallons*
1,328.41	*acre-inches*
33.21	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Roberts Ferry - 11/03/2022: Oats, silage-soft dough

Field name: Roberts Ferry Crop: Oats, silage-soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	120.05	17.05	193.59	1,103.14
Fresh water	3.94	0.00	0.00	377.87
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	130.99	17.05	193.59	1,481.01
Anticipated crop nutrient removal	165.00	26.40	136.95	0.00
Actual crop nutrient removal	109.21	19.97	112.75	719.62
Nutrient balance	21.78	-2.92	80.84	761.40
Applied to removed ratio	1.20	0.85	1.72	2.06

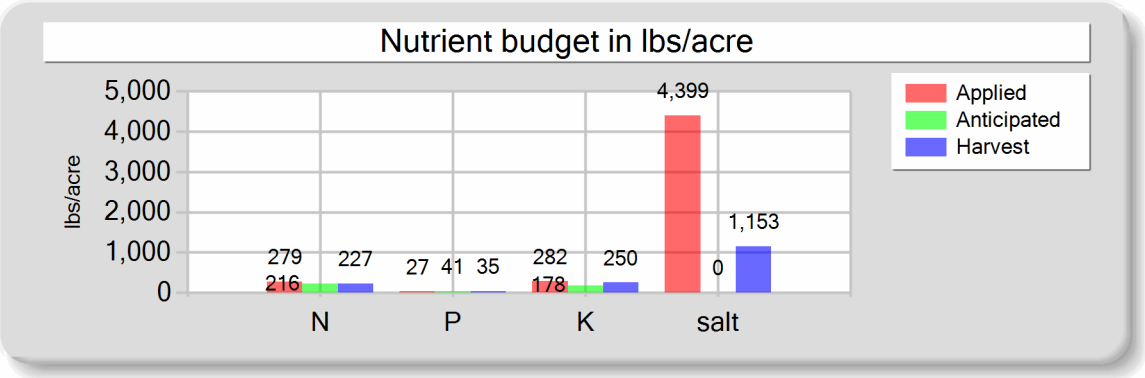
Fresh water applied	
4,830,000.00	*gallons*
177.87	*acre-inches*
4.45	*inches/acre*

Process wastewater applied	
1,728,000.00	*gallons*
63.64	*acre-inches*
1.59	*inches/acre*

Total harvests for the crop	
1	*harvests*

Roberts Ferry - 05/20/2023: Corn, silage

Field name: Roberts Ferry Crop: Corn, silage Plant date: 05/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	236.93	26.97	282.49	1,239.33
Fresh water	27.32	0.00	0.00	3,159.53
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	279.26	26.97	282.49	4,398.86
Anticipated crop nutrient removal	216.00	40.50	178.20	0.00
Actual crop nutrient removal	226.99	35.25	249.87	1,153.13
Nutrient balance	52.27	-8.28	32.62	3,245.73
Applied to removed ratio	1.23	0.77	1.13	3.81

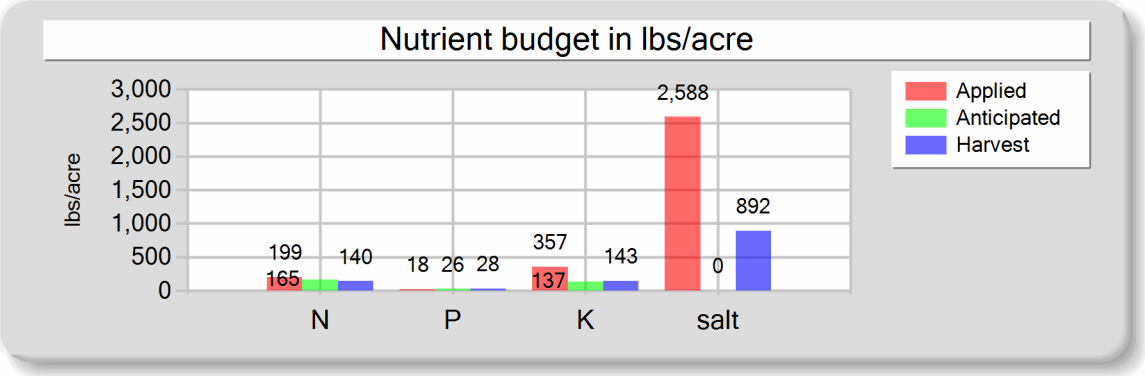
Fresh water applied	
32,499,000.00	*gallons*
1,196.83	*acre-inches*
29.92	*inches/acre*

Process wastewater applied	
4,368,000.00	*gallons*
160.86	*acre-inches*
4.02	*inches/acre*

Total harvests for the crop	
1	*harvests*

Upper Pamona - 10/27/2022: Oats, silage-soft dough

Field name: Upper Pamona | Crop: Oats, silage-soft dough | Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	180.93	18.30	356.69	2,434.34
Fresh water	11.08	0.00	0.00	153.34
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	199.01	18.30	356.69	2,587.68
Anticipated crop nutrient removal	165.00	26.40	136.95	0.00
Actual crop nutrient removal	140.07	27.95	142.59	892.45
Nutrient balance	58.94	-9.65	214.10	1,695.22
Applied to removed ratio	1.42	0.65	2.50	2.90

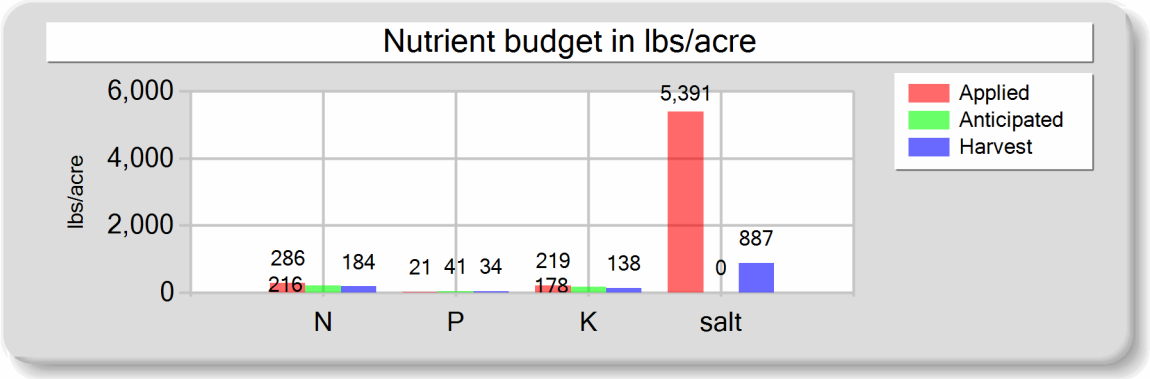
Fresh water applied	
2,250,000.00	*gallons*
82.86	*acre-inches*
2.76	*inches/acre*

Process wastewater applied	
2,256,000.00	*gallons*
83.08	*acre-inches*
2.77	*inches/acre*

Total harvests for the crop	
1	*harvests*

Upper Pamona - 06/11/2023: Corn, silage

Field name: Upper Pamona | Crop: Corn, silage | Plant date: 06/11/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	183.99	20.95	219.37	962.41
Fresh water	87.22	0.00	0.00	4,428.19
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	286.21	20.95	219.37	5,390.60
Anticipated crop nutrient removal	216.00	40.50	178.20	0.00
Actual crop nutrient removal	183.97	34.36	137.98	887.50
Nutrient balance	102.24	-13.41	81.39	4,503.10
Applied to removed ratio	1.56	0.61	1.59	6.07

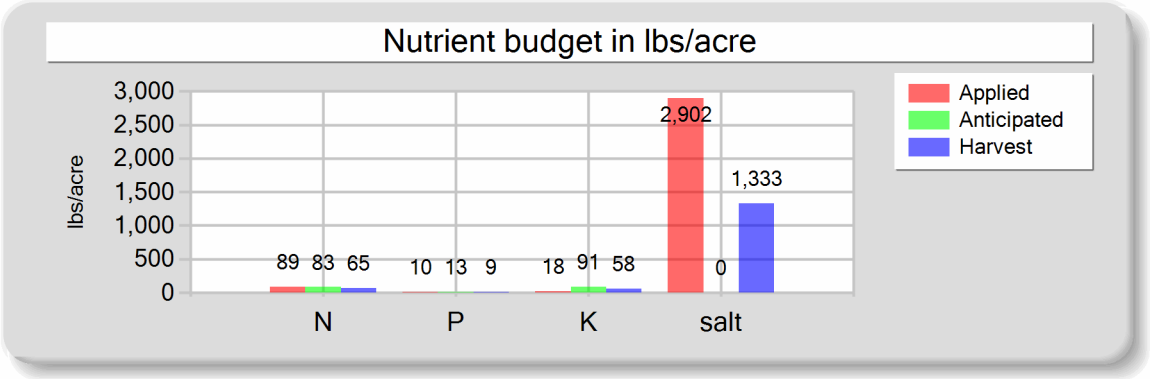
Fresh water applied	
24,120,000.00	*gallons*
888.26	*acre-inches*
29.61	*inches/acre*

Process wastewater applied	
2,544,000.00	*gallons*
93.69	*acre-inches*
3.12	*inches/acre*

Total harvests for the crop	
1	*harvests*

Vines - 02/07/2014: Grape

Field name: Vines | Crop: Grape | Plant date: 02/07/2014



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	28.80	0.00	0.00	0.00
Dry manure	45.82	10.22	17.92	1,823.97
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,077.58
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	88.62	10.22	17.92	2,901.55
Anticipated crop nutrient removal	83.00	13.00	91.00	0.00
Actual crop nutrient removal	65.12	9.20	57.62	1,332.88
Nutrient balance	23.50	1.02	-39.70	1,568.67
Applied to removed ratio	1.36	1.11	0.31	2.18

Fresh water applied	
102,716,250.00	*gallons*
3,782.69	*acre-inches*
36.03	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids

Sample date: 11/10/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 22.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,600.00	3,480.00	6,100.00							62.10
DL	100.00	100.00	100.00							0.01

Settling Basin

Sample and source description: Settling Basin

Sample date: 11/10/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 45.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,300.00	5,110.00	9,750.00							60.50
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 07/18/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 22.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,400.00	5,810.00	10,800.00							73.60
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 07/18/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 50.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,100.00	4,050.00	9,980.00							48.00
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

4QWW

Sample and source description: 4QWW

Sample date: 11/03/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	249.00	173.00	0.00	0.00	13.20	596.00								4,250.00	4,600
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

1QWW

Sample and source description: 1QWW

Sample date: 02/23/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	333.00	218.00	0.00	0.00	47.30	537.00								4,350.00	3,060
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

2QWW

Sample and source description: 2QWW

Sample date: 07/18/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	538.00	378.00	0.00	0.00	32.10	680.00								6,550.00	3,700
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

3QWW

Sample and source description: 3QWW

Sample date: 07/18/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	260.00	188.00	0.00	0.00	29.60	310.00								3,420.00	1,360
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

IW Diamond S

Fresh Water

Sample description: Fresh Water

Sample date: 10/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								170.00	132
DL	0.50	0.50	0.50								1.00	10

IW Pamona

IW Pamona

Fresh Water

Sample description: Fresh Water

Sample date: 11/01/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	13.00	0.00	5.20								963.00	660
DL	0.50	0.50	0.50								1.00	10

IW Robert

Fresh Water

Sample description: Fresh Water

Sample date: 10/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	4.03	0.00	4.01								779.00	466
DL	0.50	0.50	0.50								1.00	10

River

Fresh Water

Sample description: Fresh Water

Sample date: 10/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								82.00	64
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Almond - 02/02/2015: Almond Field Run

Almonds

Sample and source description: Almonds

Sample date: 09/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 8.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,800.00	2,380.00	37,000.00		44.20
DL	100.00	100.00	100.00		0.01

Bridge - 10/19/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 74.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,070.00	590.00	3,300.00		8.70
DL	100.00	100.00	100.00		0.01

Bridge - 05/19/2023: Corn, silage

Bridge - 05/19/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 54.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,940.00	1,010.00	4,170.00		6.20
DL	100.00	100.00	100.00		0.01

Bull Field - 10/21/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,370.00	580.00	2,950.00		7.10
DL	100.00	100.00	100.00		0.01

Bull Field - 05/20/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 52.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,740.00	1,320.00	3,760.00		4.60
DL	100.00	100.00	100.00		0.01

Dryland 1 - 10/31/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,840.00	880.00	3,960.00		6.70
DL	100.00	100.00	100.00		0.01

Encino - 11/04/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 78.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,660.00	530.00	2,040.00		6.70
DL	100.00	100.00	100.00		0.01

Encino - 06/09/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,860.00	760.00	3,240.00		6.10
DL	100.00	100.00	100.00		0.01

Indian - 10/20/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,280.00	560.00	3,180.00		7.90
DL	100.00	100.00	100.00		0.01

Indian - 05/23/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 56.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,620.00	1,190.00	3,640.00		5.20
DL	100.00	100.00	100.00		0.01

Machinery - 10/24/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,850.00	730.00	4,720.00		9.90
DL	100.00	100.00	100.00		0.01

Machinery - 06/08/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,610.00	1,060.00	6,780.00		6.70
DL	100.00	100.00	100.00		0.01

No. 4 - 10/25/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,680.00	680.00	3,140.00		8.10
DL	100.00	100.00	100.00		0.01

No. 4 - 06/14/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,680.00	720.00	2,970.00		4.00
DL	100.00	100.00	100.00		0.01

Pamona - 10/26/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,920.00	660.00	2,870.00		7.70
DL	100.00	100.00	100.00		0.01

Pamona - 06/15/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,300.00	560.00	3,220.00		5.00
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,320.00	620.00	2,490.00		5.60
DL	100.00	100.00	100.00		0.01

Roberts Ferry - 11/03/2022: Oats, silage-soft dough

Roberts Ferry - 11/03/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,390.00	620.00	3,500.00		7.30
DL	100.00	100.00	100.00		0.01

Roberts Ferry - 05/20/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,670.00	570.00	4,040.00		5.90
DL	100.00	100.00	100.00		0.01

Upper Pamona - 10/27/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 55.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,460.00	890.00	4,540.00		6.40
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

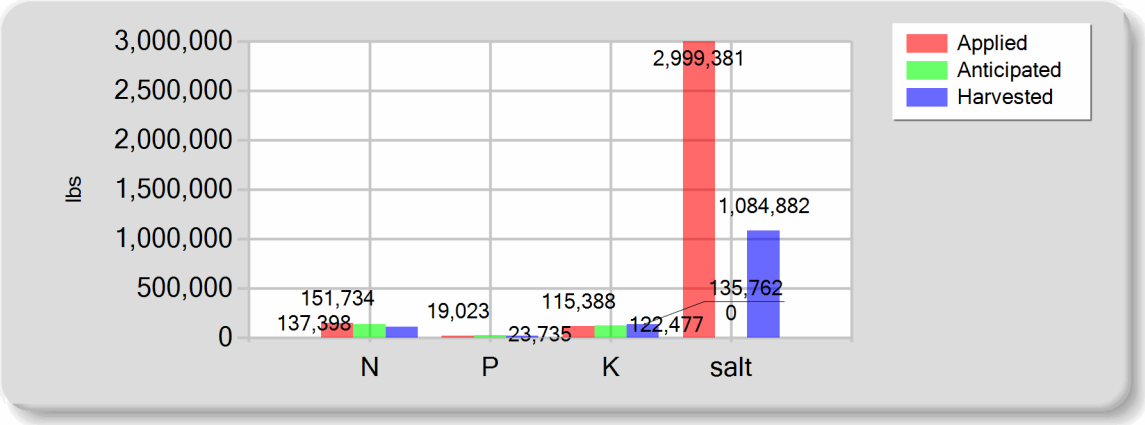
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

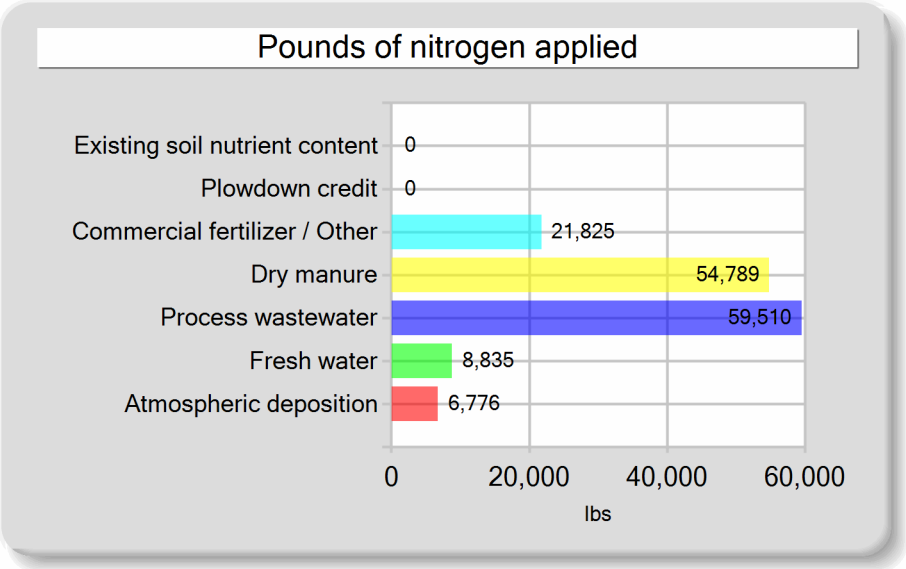
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

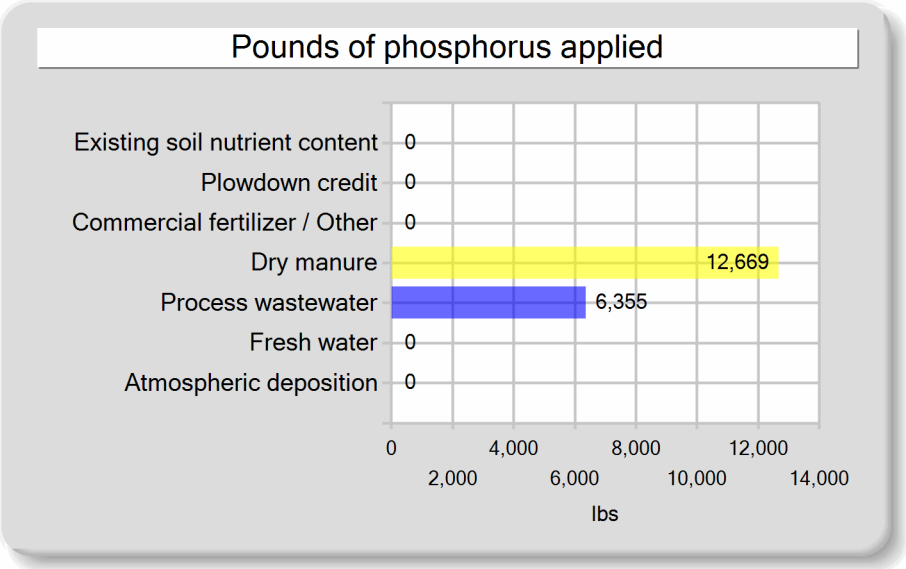
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	21,825.00	0.00	0.00	0.00
Dry manure	54,789.07	12,668.58	26,300.03	1,635,795.94
Process wastewater	59,509.60	6,354.76	89,088.15	505,482.69
Fresh water	8,834.73	0.00	0.00	858,102.25
Atmospheric deposition	6,776.00	0.00	0.00	0.00
Total nutrients applied	151,734.40	19,023.34	115,388.18	2,999,380.88
Anticipated crop nutrient removal	137,397.50	23,735.15	122,476.95	0.00
Actual crop nutrient removal	113,307.17	21,153.55	135,762.04	1,084,882.41
Nutrient balance	38,427.23	-2,130.21	-20,373.86	1,914,498.46
Applied to removed ratio	1.34	0.90	0.85	2.76

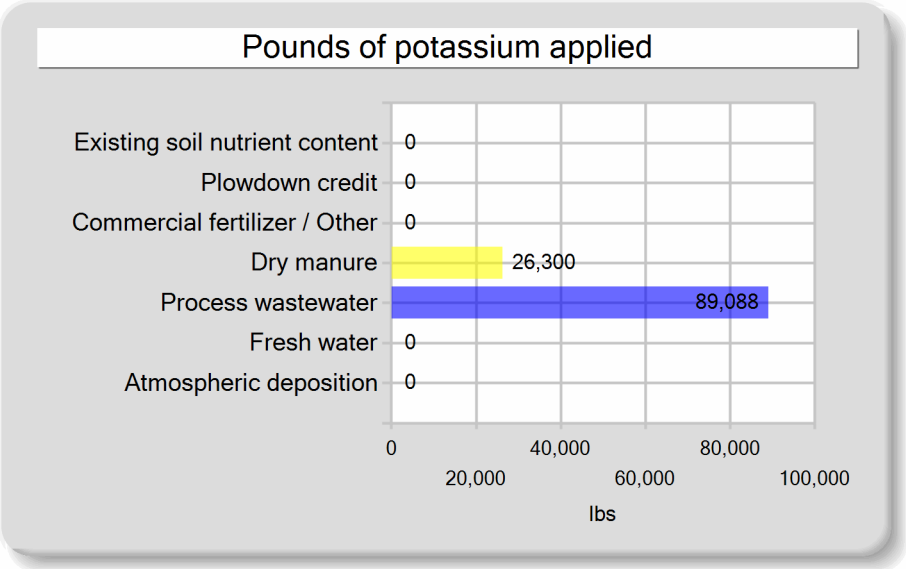
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

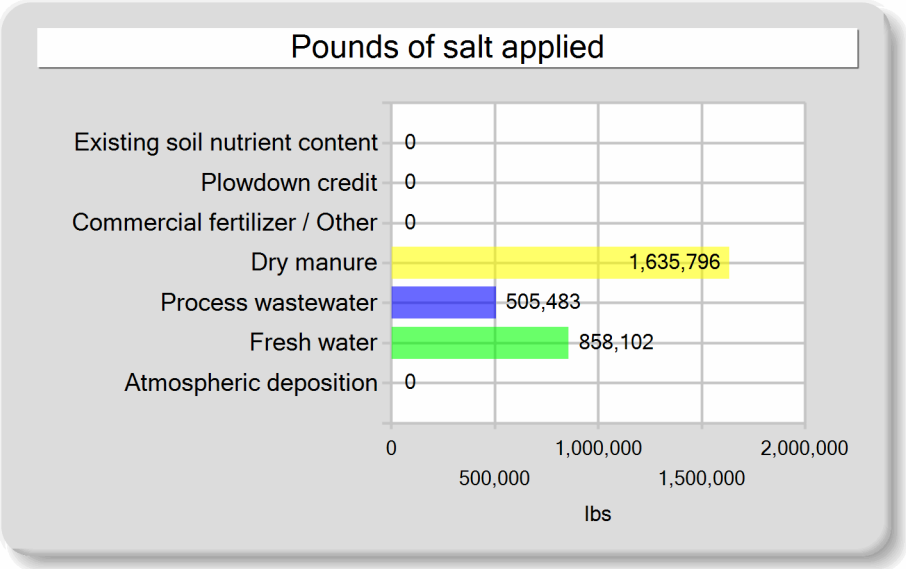


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Dryland 1 Field Oats (1.42), Pamona Field Oats (1.45), and Upper Pamona Field Oats (1.42) had slightly elevated nitrogen field ratios as fields failed to meet anticipated nitrogen removal of 165 pounds/acre. The subject fields removed 119, 127 and 140 pounds of nitrogen/acre. Yields for the fields were adequate however low nitrogen values from the plant tissue lab analysis resulted in 7.68, 7.84 and 8.92 pounds of nitrogen removed per harvested ton compared to the anticipated 10 pounds of nitrogen removed per harvested ton which ultimately led to the high nitrogen field ratios. Nitrogen application to the fields was not considered excessive but in-line with that recommended in the dairy's nutrient management plan.

Field Upper Pamona corn (1.56) had a high nitrogen field ratio as field failed to meet anticipated nitrogen removal of 216 pounds of nitrogen/acre. The subject field removed only 184 pounds of nitrogen/acre. Yield for the crop was adequate however a low nitrogen value from the plant tissue lab analysis resulted in only 6.64 pounds of nitrogen removed per harvested ton compared to the anticipated 8 pounds of nitrogen removed per harvested ton which ultimately led to the high nitrogen field ratio. Nitrogen application to the subject field was not considered excessive but in-line with that recommended in the dairy's nutrient management plan.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Chris Sawyer (May 29, 2024 18:41 PDT)	Chris Sawyer (May 29, 2024 18:41 PDT)
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Family Trust Tom Sawyer	Chris Sawyer
PRINT OR TYPE NAME	PRINT OR TYPE NAME
May 29, 2024	May 29, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

WATER WORK REQUEST

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

chase Order No

Bill To: Acct # 0 Cons # 13

Results Need By

No. Samples: No of Bottles:

Water Type: [ ] Drinking Water [ ] Wastewater
[ ] Ag Water [ ] Groundwater [ ] Monitoring Well

Other:

Name: Diamond S Ranch
Address: 600 Roberts Ferry Rd
City: Waterford **State:** CA **Zip:** 95386
Telephone: (209) 986-5562 **Fax:**
Cell/Email:
COPY TO: F&A Ag Services
& JSawyer@Fragservices.com
REQUESTED BY: Jeanette Sawyer
PROJECT:
CROP:

Analysis and Bottles Required: (*Please indicate Analysis*)

DWW1

[ ] Co. Health Dept
[ ] RWQCB [ ] Copy of Chain
[ ] State Forms [ ] QA/QC Documents

Sampled By:

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field EC
1	DW Helen	10/25/23	Started 8 AM		
2	DW LN'S	↓			
3	DW Mill Barn				
4					
5					
6					
7					
8					
9					
10					

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	Jeanette Sawyer	F&R Ag Services		
Second				
Third				
Fourth				

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.

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Invoicing Information:

Shipping

pling hrs $ In
s $ Out
Consulting

Amt Paid	Rec By	Check #	Date

Signature

Sample received in cooler with ice (coolant)
[ ] Yes [ ] No

DELLAVALLE™ LABORATORY INC

Report of Dairy Well Water Analysis

Lab No.: 23J1798
Sampled By:
Requested By: Jeanette Sawyer
Submitted Date: 10/26/23
Reported Date: 11/06/23
Project:
Crop ID:

Diamond S Ranch
600 Roberts Ferry Rd
Waterford CA 95386
00-0026269 13

E-mail: jsawyer@fragservices.com
Copy To:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	TKN mg/L	Total Nitrogen TN mg/L	pH at 25°C unit
1	DW Heifer	10/25/23	8:00	463	0.46	4.2	*		ND	4.33	7.7
2	DW LNS	10/25/23	8:00	580	0.58	1.0	*		ND	1.23	7.8
3	DW Milk Barn	10/25/23	8:00	378	0.38	2.8	*		ND	2.96	7.7

* = Field NH4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

WATER WORK REQUEST

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

Purchase Order No

Bill To: Acct # 0 Cons # 13

No. Samples: No of Bottles:

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater
[ ] Ag Water [ ] Groundwater [ ] Monitoring Well

Name: Diamond S Ranch
Address: 600 Roberts Ferry Rd
City: Waterford **State:** CA **Zip:** 95386
Telephone: 209 986-5562 **Fax:**
Cell/Email:
COPY TO: F&A Ag Services
& JSawyer@Fragservices.com
REQUESTED BY: Jeanette Sawyer
PROJECT:
CROP:

Other:

Analysis and Bottles Required: (*Please indicate Analysis*)

DCWI

[ ] Co. Health Dept
[ ] RWQCB [ ] Copy of Chain
[ ] State Forms [ ] QA/QC Documents

Sampled By:

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field EC
1	IW Diamond S Partnership	10/25/23	started 7 AM		
2	IW Roberts Ferry	↓			
3	River				
4					
5					
6					
7					
8					
9					
10					

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	Jeanette Sawyer	F&R Ag Services		
Second				
Third				
Fourth				

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Invoicing Information:

Sampling hrs
Miles
Consulting

Shipping
$ In
$ Out

Amt Paid	Rec By	Check #	Date

Signature
Sample received in cooler with ice (coolant)
[ ] Yes [ ] No

DELLAVALLE™
LABORATORY INC

Report of Dairy Canal Water Analysis

Lab No.: 23J1799
Sampled By:
Requested By: Jeanette Sawyer
Submitted Date: 10/26/23
Reported Date: 11/02/23
Project:
Crop ID:

Diamond S Ranch
600 Roberts Ferry Rd
Waterford CA 95386
00-0026269 13

E-mail: jsawyer@fragservices.com
Copy To:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	TDS mg/L	TKN mg/L	Total Nitrogen TN mg/L
1	IW Diamond S Partnership	10/25/23	7:00	170	0.17	0.7	132	ND	ND
2	IW Roberts Ferry	10/25/23	7:00	779	0.78	40.1	466	ND	4.03
3	River	10/25/23	7:00	82	0.08	0.4	63.5	ND	ND

ND = None Detected

Approved By: Scott McFrickland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

WATER WORK REQUEST

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

Chase Order No

Bill To: Acct # 0 Cons # 13

No. Samples: No of Bottles:

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater
[ ] Ag Water [ ] Groundwater [ ] Monitoring Well

Name: Diamond S Ranch

Address: 600 Roberts Ferry Rd

City: Waterford **State:** CA **Zip:** 95386

Telephone: **Fax:**

Cell/Email:

COPY TO: F&A Ag Services
& J Sawyer @ Fragservices.com

REQUESTED BY: Jeanette Sawyer

PROJECT:

CROP:

Other:

Analysis and Bottles Required: (*Please indicate Analysis*)

DCW1

[ ] Co. Health Dept

[ ] RWQCB [ ] Copy of Chain

[ ] State Forms [ ] QA/QC Documents

Sampled By:

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field EC
1	J.W Pomona	11/1/23	7 AM		
2					
3					
4					
5					
6					
7					
8					
9					
10					

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	Jeanette Sawyer	F&R Ag Services		
Second				
Third				
Fourth				

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.

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Invoicing Information:

Sampling hrs

Miles

Consulting

Amt Paid	Rec By	Check #	Date

Shipping

$ In

$ Out

Signature

Sample received in cooler with ice (coolant)

[ ] Yes [ ] No

DELLAVALLE LABORATORY INC

Report of Dairy Canal Water Analysis

Diamond S Ranch
600 Roberts Ferry Rd
Waterford CA 95386
00-0026269 13

E-mail: jsawyer@fragservices.com
Copy To:

Lab No.: 23K0165
Sampled By:
Requested By: Jeanette Sawyer
Submitted Date: 11/02/23
Reported Date: 11/14/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	TDS mg/L	TKN mg/L	Total Nitrogen TN mg/L
1	IW Pamona	11/01/23	7:00	963	0.96	5.2	660	7.6	13

ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

2023 Annual Report Diamond S

Final Audit Report 2024-05-30

Created:	2024-05-29
By:	Vince Furtado (vfurtado@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAUO1B6q6EDxVDVHnAxKKGXIrL1IvGpeuQ

"2023 Annual Report Diamond S" History

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